UNIT 4: ER MODEL

Complete Question Bank with Answers & ER Diagrams

Easy • Medium • Hard • Conversion Practice • Tutorial Solutions

Prepared from the supplied Unit4_ERmodel_final.pdf

Includes 60+ original practice questions, worked conversion examples, diagrammatic schema-to-ER answers, and solutions to the identifiable tutorial/exercise prompts in the 103-page reference.

How to Use This Question Bank

1. Attempt each question before reading its answer. 2. For conversion questions, identify keys first, then relationships, cardinality and participation. 3. In ER diagrams, rectangles are entities, diamonds are relationships, ovals are attributes, underlining marks key attributes, double lines indicate total participation, and a double diamond marks an identifying relationship. 4. When a schema does not reveal optionality or real-world rules, state a reasonable assumption. 5. The source PDF includes several worked examples and image-based diagrams; the tutorial section below covers the explicit tutorial questions and the principal exercise prompts that can be identified in the slides. It does not claim to reproduce every worked-example slide as a separate tutorial question.

Notation used below: PK = primary key; FK = foreign key; (PK) is underlined in the diagrams where practical; 1 and N label mapping cardinality. A double line means total participation. Schema names and assumptions follow the reference PDF where possible.

3. Easy Level — Fundamentals & Direct Application

20 questions • Answer and reasoning provided for every question.

E1. An entity is: (A) a table only (B) a distinguishable real-world object (C) a relationship attribute (D) a domain

Answer / reasoning: B. An entity is a real-world object distinguishable from other objects.

E2. A collection of entities of the same type is an: (A) attribute (B) entity set (C) domain (D) key

Answer / reasoning: B. An entity set groups entities that share the same properties.

E3. The set of permitted values of an attribute is its: (A) degree (B) domain (C) participation (D) cardinality

Answer / reasoning: B. A domain defines allowable values.

E4. Which is usually a derived attribute? (A) date_of_birth (B) ID (C) age computed from date_of_birth (D) email

Answer / reasoning: C. Age can be calculated from date of birth.

E5. An attribute with components such as first name and last name is: (A) simple (B) composite (C) derived (D) keyless

Answer / reasoning: B. A composite attribute has meaningful components.

E6. Which is multivalued? (A) birth place (B) salary (C) phone numbers for one person (D) course ID

Answer / reasoning: C. One person may have several phone numbers.

E7. In Chen notation, a relationship set is shown by a: (A) rectangle (B) diamond (C) oval (D) dashed oval

Answer / reasoning: B. Diamonds represent relationship sets.

E8. A primary key attribute is commonly shown by: (A) underline (B) double rectangle (C) double diamond (D) dashed line

Answer / reasoning: A. Underlining identifies key attributes.

E9. A relationship involving two entity sets is: (A) unary (B) binary (C) ternary (D) weak

Answer / reasoning: B. Degree two means binary.

E10. A relationship involving three entity sets is: (A) binary (B) ternary (C) recursive only (D) derived

Answer / reasoning: B. Ternary relationships involve three entity sets.

E11. A super key is a set of attributes that: (A) may not identify rows (B) uniquely identifies an entity (C) is always minimal (D) must be a foreign key

Answer / reasoning: B. A super key uniquely identifies an entity; it need not be minimal.

E12. A minimal super key is called a: (A) domain (B) candidate key (C) descriptive attribute (D) role

Answer / reasoning: B. Candidate keys have no unnecessary attribute.

E13. The chosen candidate key is the: (A) partial key (B) foreign key (C) primary key (D) domain

Answer / reasoning: C. One candidate key is selected as the primary key.

E14. An entity set with sufficient attributes to form a primary key is: (A) weak (B) strong (C) derived (D) multivalued

Answer / reasoning: B. A strong entity set has its own key.

E15. The partial key is associated with a: (A) weak entity (B) domain (C) relationship degree (D) derived attribute

Answer / reasoning: A. A partial key distinguishes weak entities under the same owner.

E16. Total participation means: (A) no entity participates (B) every entity participates at least once (C) every entity has many keys (D) relationship is ternary

Answer / reasoning: B. Every entity in the set must participate.

E17. Partial participation means: (A) some entities may not participate (B) all must participate (C) entity is weak (D) relationship has no attributes

Answer / reasoning: A. Participation is optional for some entities.

E18. Which relationship is many-to-many? (A) each student has one ID (B) students take many subjects and subjects have many students (C) each employee has one department (D) one institute has one head

Answer / reasoning: B. Both sides can associate with many entities.

E19. The role describes: (A) an entity's function in a relationship (B) a domain (C) a key constraint (D) a table name

Answer / reasoning: A. Role names distinguish how an entity set participates.

E20. A relationship descriptive attribute belongs to: (A) always one entity (B) the association itself when its value depends on that association (C) only a weak entity (D) a domain

Answer / reasoning: B. Example: advisor date is a property of an instructor–student association.

4. Medium Level — Applied Concepts

20 questions • Answer and reasoning provided for every question.

M1. A customer may own many cars; each car has one owner. Cardinality? (A) 1:1 (B) 1:N customer-to-car (C) M:N (D) ternary

Answer / reasoning: B. One customer can relate to several cars, while each car has one owner under the stated rule.

M2. Every loan must belong to at least one customer, but a customer may have no loan. Participation? (A) customer total, loan partial (B) both partial (C) loan total, customer partial (D) both total

Answer / reasoning: C. Loan cannot exist without a customer; customers may have no loans.

M3. Employee(EmpNo, Name, DeptNo), Department(DeptNo, Name). What does DeptNo in Employee usually represent? (A) multivalued attribute (B) FK to Department (C) derived key (D) relationship degree

Answer / reasoning: B. In a many-employees-to-one-department relationship, the FK is stored on the many side.

M4. Student and Subject: students take many subjects and subjects have many students. Best schema mapping? (A) FK only in Student (B) FK only in Subject (C) separate Takes(RollNo, SubjectID) (D) merge entities

Answer / reasoning: C. M:N is represented by a separate relation with both entity keys.

M5. An instructor can advise zero or many students; every student has exactly one advisor. Which limits? (A) instructor 1..1, student 0..* (B) instructor 0..*, student 1..1 (C) both 0..* (D) both 1..1

Answer / reasoning: B. This matches the source PDF's advisor example.

M6. For weak entity Section owned by Course, partial key is Sec_ID and other key parts are semester/year. Primary key includes: (A) Sec_ID only (B) course key plus discriminator attributes (C) course title only (D) instructor key

Answer / reasoning: B. The owner key is combined with the weak entity discriminator.

M7. A person has first, middle, last name. When mapped to a relational schema, composite name is usually: (A) stored as one required composite column (B) flattened into component columns (C) always multivalued (D) dropped

Answer / reasoning: B. The source says composite attributes are flattened into components.

M8. Instructor has multiple phone numbers. Appropriate mapping? (A) put all phones in one atomic cell (B) separate InstructorPhone(ID, Phone) relation (C) create one instructor per phone (D) use phone as PK of Instructor

Answer / reasoning: B. A separate relation stores one tuple per phone value.

M9. In a recursive WorksFor(Employee, Manager) relationship, the manager reference can be represented as: (A) manager_id FK referencing Employee(employee_id) (B) a new entity type always (C) a domain only (D) no key

Answer / reasoning: A. Both roles use the same entity set; role-specific FK points back to Employee.

M10. A relationship is ternary when: (A) three attributes exist (B) three entity sets participate (C) it has three keys (D) participation is total

Answer / reasoning: B. Degree counts participating entity sets, not attributes.

M11. If (ID, Name) uniquely identifies Instructor but ID alone is unique, then (ID, Name) is: (A) candidate key only (B) super key but not minimal (C) not a super key (D) foreign key

Answer / reasoning: B. ID is a proper subset that already uniquely identifies the instructor.

M12. An advisor relationship has attribute Advisor_On_Date. Under the source's described relationship-key rule, the full relationship instance can be distinguished by: (A) date alone always (B) entity keys plus descriptive attribute (C) student name only (D) instructor salary

Answer / reasoning: B. The PDF describes entity keys union descriptive attributes for relationship instances.

M13. A department offers many courses and each course belongs to one department. Schema mapping typically places Department key: (A) in Department as FK to Course (B) in Course as FK to Department (C) in both without reason (D) nowhere

Answer / reasoning: B. The one-side key is added to the many-side relation.

M14. Which ER notation denotes a weak entity? (A) single oval (B) double rectangle (C) dashed oval only (D) arrow

Answer / reasoning: B. Weak entity sets are shown with double rectangles.

M15. Which is the identifying relationship notation? (A) double diamond (B) dashed line (C) single oval (D) triangle

Answer / reasoning: A. The identifying relationship is drawn as a double diamond.

M16. In a 1:1 relationship, when mapping to relations, the FK may be placed: (A) on either side, considering participation (B) only on the alphabetically first entity (C) only in a third table always (D) nowhere

Answer / reasoning: A. Either side can be chosen; total participation and nulls guide the choice.

M17. A project is funded by many agencies and an agency funds many projects. Grant amount depends on a particular project-agency pair. Grant amount is best modeled as: (A) an attribute of Agency only (B) an attribute of Project only (C) a relationship attribute on Funds (D) a primary key necessarily

Answer / reasoning: C. It describes the funding association.

M18. A derived age attribute is usually not stored redundantly because: (A) it is always a key (B) it can be computed from date of birth (C) it is multivalued (D) it is a weak key

Answer / reasoning: B. Storing a derived value can become inconsistent.

M19. Which case is total participation? (A) some instructors may advise nobody (B) every loan must be associated with customer(s) (C) a customer may have no loan (D) optional phone number

Answer / reasoning: B. Every loan must take part in Borrower.

M20. A relation schema shows Student(RollNo, Name), Subject(SubjectID, Title), Takes(RollNo, SubjectID). The best inferred relationship is: (A) 1:1 (B) M:N (C) recursive (D) weak ownership

Answer / reasoning: B. A separate relation containing both keys generally maps a many-to-many association.

5. Hard Level — Multi-step Reasoning

20 questions • Answer and reasoning provided for every question.

H1. A company schema has Employee(EmpID, Name, DeptID, ManagerID), Department(DeptID, Name), Project(ProjectID, Title), WorksOn(EmpID, ProjectID, Hours). Infer relationships and cardinalities.

> **Answer / reasoning:** Employee–Department is N:1 if each employee has at most one department; ManagerID is a recursive Employee-to-Employee FK; Employee–Project is M:N through WorksOn, with Hours as relationship attribute.

H2. A schema has Order(OrderID, CustomerID, Date), OrderLine(OrderID, LineNo, ProductID, Qty). Identify the weak entity and key under the usual assumption that line numbers are unique only within an order.

> **Answer / reasoning:** OrderLine is weak/existence-dependent on Order; partial key LineNo; PK (OrderID, LineNo); ProductID references Product. OrderLine total participation in its identifying Order relationship.

H3. A relation Advisor(InstID, StudentID, StartDate) has PK (InstID, StudentID). What does the key tell you, and what if multiple advisor periods per student–instructor pair must be stored?

> **Answer / reasoning:** The given key permits only one row per pair. If multiple periods are allowed, include a distinguishing attribute such as StartDate in the PK (InstID, StudentID, StartDate), assuming dates distinguish periods.

H4. You see Department(DeptID, Name) and Employee(EmpID, DeptID, Name). Does the schema prove every department has an employee? Explain.

> **Answer / reasoning:** No. Employee.DeptID indicates employee-to-department association, but it does not enforce every department participating at least once. That is a business rule/participation constraint not inferable from this schema alone.

H5. A hospital stores Patient(PID), Doctor(DID), Treatment(PID,DID,Date,Outcome). Each patient may see many doctors and each doctor treats many patients. Explain mapping and why Date may be part of the key.

> **Answer / reasoning:** Treatment maps the M:N relationship. If each patient-doctor pair can have multiple treatment dates, PK should include Date: (PID,DID,Date); Outcome is descriptive relationship data.

H6. A weak entity Dependent has Name as partial key and belongs to Employee(EmpID). Can Name alone be its PK?

> **Answer / reasoning:** No, because different employees may have dependents with the same name. The full PK is (EmpID, Name), assuming a dependent name is unique within one employee's dependents.

H7. A student can enroll in a course multiple times in different semesters. Is Enroll(StudentID, CourseID) sufficient?

> **Answer / reasoning:** No, if repeated enrollments are permitted. Add Semester/Year (or another attempt identifier) to distinguish each enrollment; e.g. PK (StudentID, CourseID, Semester, Year).

H8. An agency funds many projects; each project must receive one or more grants. Agencies may exist before funding a project. State cardinality and participation.

> **Answer / reasoning:** Agency–Project is M:N. Project participation in Funds is total (minimum one agency); Agency participation is partial (may fund zero projects). Grant attributes such as GrantOrderNo/amount belong to the relationship if they describe a particular funding event.

H9. A schema uses Employee(EmpID, DeptID) and also Department(DeptID, ManagerID). What relationship is represented by ManagerID?

> **Answer / reasoning:** It represents Department–Employee manages relationship, typically one manager employee per department if ManagerID is a single FK; multiple departments could reference the same employee unless a uniqueness constraint says otherwise.

H10. A model says each book has one author and one publisher, but authors can write many books. A book can have several categories and reviews by authors. Propose entities and relationships.

> **Answer / reasoning:** Book(ISBN, title, year, page_count, price), Author(AuthorID,...), Publisher(PublisherID,...), Edition(ISBN, EditionNo, Type) if editions are identified per book; Writes is 1:N from Author to Book under the stated

assumption; PublishedBy is N:1 Book-to-Publisher; Category is multivalued or a Category entity; Review is an Author–Book relationship with Rating/Feedback attributes.

H11. A ternary Supply(Supplier, Part, Project, Qty) is decomposed into three binary relationships. Why might this change the meaning?

Answer / reasoning: The quantity applies to the exact supplier-part-project combination. Three binary relationships may allow combinations that never existed in the original ternary association; retain a ternary relationship/relation unless constraints prove decomposition lossless.

H12. An entity has candidate keys Email and EmployeeID. Which is PK? Must Email be discarded?

Answer / reasoning: Choose the key that best fits the entity semantics and stability (often EmployeeID). Email remains an alternate candidate key and may still have a uniqueness constraint.

H13. A one-to-many relationship is mapped by placing DeptID in Employee. What can go wrong if the relationship is optional for employees?

Answer / reasoning: DeptID may need to be nullable for employees not assigned to a department. If every employee must belong to a department, Employee participation is total and DeptID should be required.

H14. A schema includes Section(CourseID, SecID, Semester, Year, Room). Why is a separate Sec_Course(CourseID, SecID, Semester, Year) relation redundant under the source mapping rule?

Answer / reasoning: Section already contains the identifying Course key and weak entity discriminator, so the identifying relationship is represented by those columns; a duplicate relation repeats the same association.

H15. For a recursive employee hierarchy, what is the difference between ManagerID as an FK and a separate Manager entity set?

Answer / reasoning: Manager is a role played by Employee, not necessarily a distinct entity type. ManagerID references Employee(EmpID), and role names distinguish manager from subordinate.

H16. A customer can own one or more cars; each car may have zero or many accidents recorded. Identify what is known and one ambiguity.

Answer / reasoning: Customer-to-Car is 1:N, with Customer total participation if every customer must own at least one car. Car-to-Accident is ambiguous because an accident can involve multiple cars; model Accident as entity and a potentially M:N InvolvedIn relationship unless the requirement says each accident belongs to one car.

H17. A relation Enrolls(StudentID, CourseID, Grade) has a unique key only on StudentID. Is that consistent with many-to-many enrollment?

Answer / reasoning: Not if students can enroll in multiple courses; StudentID alone would restrict each student to one enrollment. Use composite key (StudentID, CourseID), or add semester/year if repeated enrollment is allowed.

H18. A Department has composite address (Street, City, State) and multivalued phone. Show schema mapping.

Answer / reasoning: Department(DeptID, Name, Street, City, State); DepartmentPhone(DeptID, Phone) with PK (DeptID, Phone) and DeptID FK to Department. Composite components are flattened; multivalued phone is separate.

H19. An entity set participates totally in a relationship. Does that automatically make it weak?

Answer / reasoning: No. Total participation is a participation constraint; weak/strong classification depends on whether the entity set has its own sufficient key. A strong entity can participate totally.

H20. A 1:1 relationship has total participation on Institute and partial participation on Head. Where is a sensible FK?

Answer / reasoning: Put Institute's key in Head as FK if each Head belongs to one Institute; but because Institute participation is total, putting HeadID in Institute could avoid nulls. The exact choice depends on which side is existence-dependent and desired constraints.

6. ER-to-Relational-Schema Conversion Practice

Mapping rules from the reference: strong entity → relation with its attributes; weak entity → relation including owner PK plus partial key; M:N → separate relation with both participating keys and relationship attributes; 1:N → place the one-side PK as FK on the many side; 1:1 → place an FK on either side, choosing based on participation; composite attributes are flattened; multivalued attributes use a separate relation; the identifying relationship for a weak entity is usually redundant as a separate relation.

6.1 Course and Section

Task: Convert the described ER design to a relational schema. **Result:** Course(CourseID, Title, Credits); Section(CourseID, SecID, Semester, Year, Room)



Chen-style schematic; key attributes and participation are explained in the answer.

Solution: PK Course: CourseID. Section is weak; partial key is (SecID, Semester, Year). PK Section=(CourseID, SecID, Semester, Year); CourseID is FK to Course. Course 1:N Section; Section total participation because a section cannot exist without a course.

6.2 Student and Subject

Task: Convert the described ER design to a relational schema. **Result:** Student(RollNo, Name); Subject(SubjectID, Title, Credits); Takes(RollNo, SubjectID, Grade)

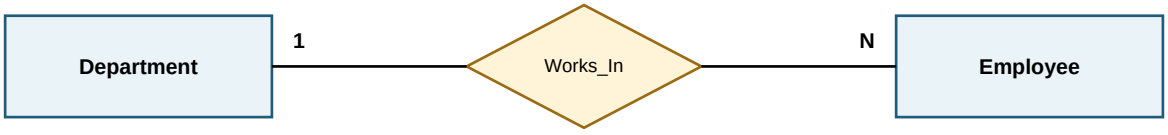


Chen-style schematic; key attributes and participation are explained in the answer.

Solution: Student and Subject are strong entities. Takes is M:N, PK=(RollNo, SubjectID) if one enrollment per pair; both are FKs. Grade belongs to Takes because it depends on the student-subject pairing.

6.3 Department and Employee

Task: Convert the described ER design to a relational schema. **Result:** Department(DeptID, DeptName); Employee(EmpID, Name, Salary, DeptID)

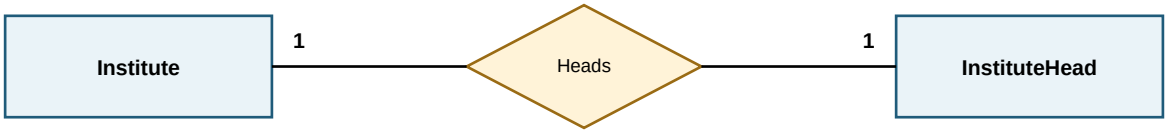


Chen-style schematic; key attributes and participation are explained in the answer.

Solution: Department PK=DeptID; Employee PK=EmpID. DeptID in Employee is FK to Department. This represents 1:N Department-to-Employee (many employees belong to at most one department). If every employee must have a department, Employee participation is total; whether every department must have employees is not implied.

6.4 Institute and Head

Task: Convert the described ER design to a relational schema. **Result:** Institute(InstName, Location, HeadName); InstituteHead(HeadName, Qualification, Experience)



Chen-style schematic; key attributes and participation are explained in the answer.

Solution: Assuming each institute has at most one head and each head heads at most one institute, the relationship is 1:1. The example schema places HeadName in Institute as FK to InstituteHead. If every institute must have a head, make that FK mandatory; avoid duplicate head details in both relations.

6.5 Employee and Project

Task: Convert the described ER design to a relational schema. **Result:** Employee(EmpID, Name); Project(ProjectID, Title); WorksOn(EmpID, ProjectID, Hours)



Chen-style schematic; key attributes and participation are explained in the answer.

Solution: WorksOn maps M:N. PK=(EmpID, ProjectID) if each employee-project pair occurs once. Both attributes are FKs. Hours is a relationship attribute because it describes the employee's work on that project.

6.6 Employee and Dependents

Task: Convert the described ER design to a relational schema. **Result:** Employee(EmpNo, EName); Dependent(EmpNo, DepName, Relationship, Age)



Chen-style schematic; key attributes and participation are explained in the answer.

Solution: Dependent is weak. EmpNo is FK to Employee. Assume DepName distinguishes dependents only within one employee; PK=(EmpNo, DepName). Employee-to-Dependent is 1:N and Dependent has total participation in identifying Depends_On.

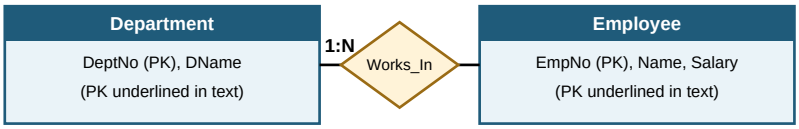
7. Relational-Schema-to-ER-Model Practice (with Drawn Diagrams)

Read each schema, infer relationships from foreign keys and relation structure, then compare your ER diagram with the drawn answer. A schema can reveal keys and links, but optionality and some business rules may require assumptions.

7.1 Given schema: `Employee(EmpNo, EName, Salary, DeptNo); Department(DeptNo, DName)`

Required: identify entities, keys, relationships, cardinality, participation, and draw an ER model.

ER model



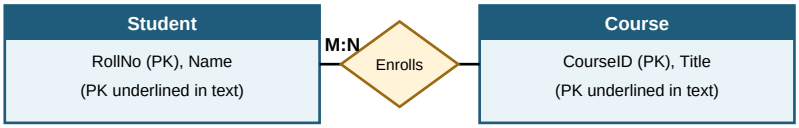
Schematic ER diagram; assumptions and participation constraints are stated in the solution.

Solution and reasoning: Employee PK=EmpNo; Department PK=DeptNo; Employee.DeptNo is FK → Department.DeptNo. Relationship Works_In is N:1 from Employee to Department (equivalently Department 1:N Employee). Assume each employee belongs to one department; if DeptNo is mandatory, Employee participation is total. Department participation is unknown from schema alone.

7.2 Given schema: `Student(RollNo, Name); Course(CourseID, Title, Credits); Enrolls(RollNo, CourseID)`

Required: identify entities, keys, relationships, cardinality, participation, and draw an ER model.

ER model



Schematic ER diagram; assumptions and participation constraints are stated in the solution.

Solution and reasoning: Student PK=RollNo; Course PK=CourseID; Enrolls has composite PK (RollNo, CourseID) and both are FKs. This indicates M:N Takes/Enrolls. Participation on either side is not guaranteed by the schema alone.

7.3 Given schema: `Employee(EmployeeID, Name, ManagerID)`

Required: identify entities, keys, relationships, cardinality, participation, and draw an ER model.



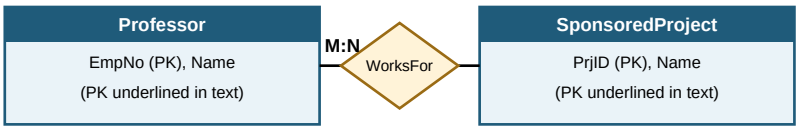
Chen-style schematic; key attributes and participation are explained in the answer.

Solution and reasoning: Employee PK=EmployeeID; ManagerID is a self-referencing FK → Employee.EmployeeID. Draw a recursive Works_For relationship with roles Manager and Worker. Each worker has at most one manager under this single-column FK; a manager may supervise many workers. Null ManagerID can represent a top-level employee.

7.4 Given schema: `Professor(EmpNo, Name, Address); SponsoredProject(PrjID, Name, Duration); WorksFor(EmpNo, PrjID, PercentTime)`

Required: identify entities, keys, relationships, cardinality, participation, and draw an ER model.

ER model



Schematic ER diagram; assumptions and participation constraints are stated in the solution.

Solution and reasoning: Professor PK=EmpNo; SponsoredProject PK=PrjID; WorksFor has composite PK (EmpNo, PrjID), FKs to both entities, and PercentTime as a relationship attribute. Relationship is M:N; participation is not inferable as total from this schema alone.

7.5 Given schema: `Employee(EmpNo, EName); Medicare_Dependents(EmpNo, Depnd_Name, Relationship_with)`

Required: identify entities, keys, relationships, cardinality, participation, and draw an ER model.



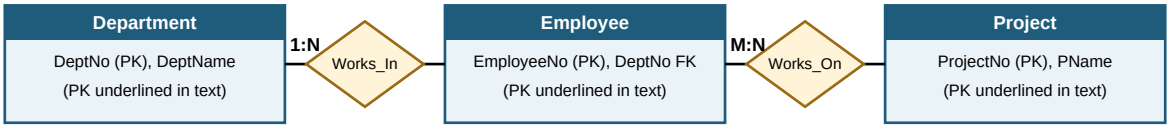
Chen-style schematic; key attributes and participation are explained in the answer.

Solution and reasoning: Employee PK=EmpNo. Medicare_Dependents PK=(EmpNo, Depnd_Name) under the assumption dependent name is unique per employee; EmpNo is FK. Model Dependents as a weak entity with partial key Depnd_Name and identifying Depends_On relationship; 1:N Employee-to-Dependents; dependent participation total.

7.6 Given schema: `Department(DeptNo, DeptName); Project(ProjectNo, PName, Budget); Employee(EmployeeNo, LName, FName, DeptNo); Works_On(ProjectNo, EmployeeNo, EnterDate, Job)`

Required: identify entities, keys, relationships, cardinality, participation, and draw an ER model.

ER model



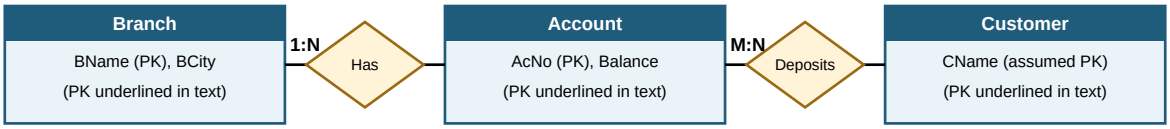
Schematic ER diagram; assumptions and participation constraints are stated in the solution.

Solution and reasoning: Employee.DeptNo FK → Department; Department-to-Employee is 1:N. Works_On maps M:N Employee–Project with EnterDate and Job as relationship attributes. If an employee can work on a project multiple times, include EnterDate in the key; otherwise PK=(ProjectNo, EmployeeNo) follows the simpler source-style mapping.

7.7 Given schema: `Account(AcNo, Balance, BName); Branch(BName, BCity, Assets); Customer(CName, CCity, CStreet); Loan(LNo, Amount, BName); Depositor(AcNo, CName); Borrower(CName, LNo)`

Required: identify entities, keys, relationships, cardinality, participation, and draw an ER model.

ER model



Schematic ER diagram; assumptions and participation constraints are stated in the solution.

Solution and reasoning: Account PK=AcNo; Branch PK=BName; Loan PK=LNo; Customer key is not explicitly marked in the source schema, so assume CName is key only for this exercise. Account.BName and Loan.BName are FKs to Branch. Depositor maps Customer–Account M:N; Borrower maps Customer–Loan M:N. Branch 1:N Account and Branch 1:N Loan.

7.8 Given schema: `Professor(Name, Phone, Address, Sex, CourseID); Course(CourseID, Name, Credits)`

Required: identify entities, keys, relationships, cardinality, participation, and draw an ER model.



Chen-style schematic; key attributes and participation are explained in the answer.

Solution and reasoning: Course PK=CourseID; Professor.CourseID is FK to Course. Under the given schema, multiple professors can reference one course and each professor references at most one course: Course 1:N Professor. If a professor may teach multiple courses, this schema is insufficient and needs a separate teaching relation.

8. Original Tutorial Questions and Exercise Solutions from the Reference PDF

This section separates source-based tutorial/exercise prompts from newly written practice questions. The reference is 103 pages. The explicit five-question tutorial at the end (PDF page 103) is reproduced in meaning below. Exercise prompts and conversion examples identified in the slides (notably pages 61–74 and 82–94) are also solved. Some pages are diagram-only worked examples rather than separately worded questions; their identifiable requirements are represented here.

Tutorial Q1 (source PDF page 103): Explain the difference between a weak and a strong entity set.

Answer: A strong entity set has attributes that can form its own primary key. A weak entity set does not have enough attributes for a standalone key and depends on an identifying strong entity. Example: Employee(EmpNo) is strong; Dependent(Name, Relationship, Age) may be weak if Name is only unique per employee. The weak entity's full key becomes (EmpNo, Name). In Chen notation, weak entities use a double rectangle and the identifying relationship a double diamond.

Tutorial Q2 (source PDF page 103): Distinguish primary key, candidate key, and super key with an example.

Answer: Suppose Instructor(ID, Name, Email), where ID and Email are each unique. Super key: any attribute set that uniquely identifies an instructor, such as {ID}, {Email}, or {ID, Name}. Candidate key: a minimal super key, so {ID} and {Email}. Primary key: the chosen candidate key, e.g. ID. {ID, Name} is not a candidate key because ID alone already identifies the row.

Tutorial Q3 (source PDF page 103): A weak entity can be made strong by adding its identifying entity's primary-key attributes. What redundancy results?

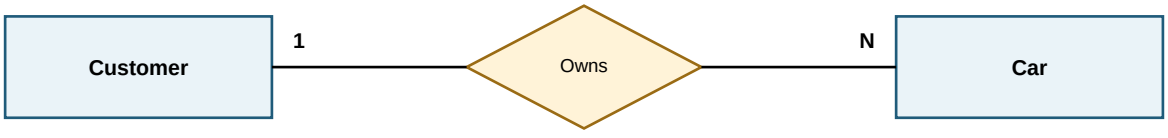
Answer: Adding the owner's key makes the weak entity's combined key unique, but it does not conceptually remove existence dependence. If the owner key is also stored redundantly in a separate identifying-relationship relation, the same association is repeated. The source notes that Section already contains CourseID, so a separate Sec_Course(CourseID, SecID, Semester, Year) relation is redundant. Keep the owner key in the weak-entity relation and do not create an unnecessary duplicate relationship relation.

Tutorial Q4 (source PDF page 103): Must an entity be weak to have total participation? Give a strong entity with total participation.

Answer: No. Weak/strong describes key sufficiency; total/partial describes participation. Example: Loan(LoanNo, Amount) can be a strong entity with its own LoanNo key, yet every loan must participate in Borrower with at least one customer. Therefore Loan is strong but total in Borrower.

Tutorial Q5 (source PDF page 103): Draw an ER model for a car-insurance company whose customers own one or more cars; each car has zero to any number of recorded accidents.

Answer: Entities: Customer(CustomerID, Name, Address), Car(RegistrationNo, Make, Model), Accident(AccidentID, Date, Location). Customer–Owns–Car is 1:N; if the statement means every customer owns at least one car, Customer participation is total and Car participation is total for ownership (each car has an owner). Car–InvolvedIn–Accident is M:N if one accident can involve several cars; Car participation in accident records is partial because a car can have zero accidents. The wording does not explicitly say whether one accident can involve multiple cars, so M:N is the safer assumption; if each accident is linked to exactly one car, use 1:N instead.



Chen-style schematic; key attributes and participation are explained in the answer.



Chen-style schematic; key attributes and participation are explained in the answer.

Exercise: Banking database ER model (source PDF page 61).

The source diagram contains Account(account-number, balance), Branch(branch-name, branch-city, assets), Customer(customer-name, customer-street, customer-city), and Loan(loan-number, amount). Relationship sets are account-branch, loan-branch, depositor, and borrower. The diagram shows Account associated with one Branch and a Branch with many Accounts; similarly, a Branch can have many Loans. Depositor connects Customer and Account, and Borrower connects Customer and Loan; these are conventionally M:N because joint accounts and joint borrowers are possible. The Account and Loan sides show total participation in their branch relationships in the source diagram. The relational schema shown later (PDF page 91) is: Account(Ac_no, Balance, B_name), Branch(B_name, B_city, Assets), Customer(C_name, C_city, C_street), Loan(L_no, Amt, B_name), Depositor(Ac_no, C_name), Borrower(C_name, L_no). B_name in Account/Loan references Branch; Depositor and Borrower contain the keys of the participating entities.

Exercise: University database ER model (source PDF page 62).

Entities include Department(dept_name, building, budget), Instructor(ID, name, salary), Student(ID, name, tot_cred), Course(course_id, title, credits), Section(sec_id, semester, year), Classroom(building, room_number, capacity), and Time_Slot(time_slot_id, day, start_time, end_time). Relationships shown include course_dept, inst_dept, stud_dept, advisor, teaches, takes (with Grade), sec_course (identifying relationship), sec_class, sec_time_slot, and recursive prereq between Course entities. Use Department 1:N Course/Instructor/Student under the depicted arrows; Student–Instructor advisor follows the slide's 1:1 maximum constraint example where each student has exactly one advisor and an instructor can advise 0..* students; Student–Section takes is M:N with Grade as relationship attribute. Section is weak under Course with composite PK (course_id, sec_id, semester, year). Prereq is recursive many-to-many. Schema examples: Course(Course_ID, Title, Credits); Section(Course_ID, Sec_ID, Semester, Year, ...); Takes(Student_ID, Section key attributes, Grade); Instructor and Department use FK placement on the many side. The diagram's arrows and double lines govern exact constraints; where slide resolution leaves a limit unclear, state the assumptions above.

Exercise: Company database ER model (source PDF page 63).

Entities: EMPLOYEE(Ssn, composite Name = Fname/Minit/Lname, Bdate, Address, Salary, Sex), DEPARTMENT(Number, Name, multivalued Locations, derived Number_of_employees), PROJECT(Number, Name, Location), and weak DEPENDENT(Name partial key, Sex, Birth_date, Relationship). Relationships: WORKS_FOR is N:1 Employee-to-Department; MANAGES is 1:1 with Start_date as a relationship attribute; CONTROLS is Department 1:N Project; WORKS_ON is M:N Employee–Project with Hours; SUPERVISION is recursive Employee-to-Employee with Supervisor/Supervisee roles; DEPENDENTS_OF is the identifying relationship, Department? [correction: Employee] to Dependent, 1:N, with total participation of Dependent. Multivalued Department Locations maps to Department_Location(DeptNo, Location). Derived Number_of_employees is calculated rather than stored. Dependent PK = (Employee Ssn, Dependent Name).

Exercise: Funded-project institute scenario (source PDF pages 65–69).

Entities and attributes: Department(DeptNo, DName), Faculty(EmpNo, EName, Qualification, ResearchDomain), Project(ProjID, ProjName, Duration, FundReceived), Agency(GrantOrderNo, AgencyName, ContactPerson, Email, Phone, TotalGrant, GrantYear). The source states many faculties work in each department; each department may have many projects; agencies and projects are M:N because an agency can fund multiple projects and a project can receive grants from multiple agencies. Model Department–Faculty as 1:N, Department–Project as 1:N, Project–Agency as M:N with relationship name Granted_By. Qualification may be multivalued because examples list multiple

qualifications. If grant-order number identifies each grant transaction, treat it as an Agency/grant identifier or a relationship attribute/entity according to the real-world rule; the slide lists it as an Agency attribute. Participation: Project in Granted_By is total because each project is funded by one or more agencies; Agency may be partial unless every agency must fund a project. A plausible schema is Department(DeptNo PK,DName); Faculty(EmpNo PK,EName,DeptNo FK); FacultyQualification(EmpNo FK,Qualification); Project(ProjID PK,ProjName,Duration,FundReceived,DeptNo FK); Agency(GrantOrderNo PK,AgencyName,ContactPerson,Email,Phone,TotalGrant,GrantYear); ProjectAgency(ProjID FK,GrantOrderNo FK) with composite PK unless multiple grant transactions require another key.

Exercise: Online book database (source PDF pages 70–74).

Book attributes: ISBN, Title, Category (multivalued in the slide), Price, Year, PageCount. Author: A_ID, Name, Phone (phone shown multivalued). Publisher: PubID, Name, PhoneNo (multivalued), EmailId. Edition: Edition_No, Type; the slide does not clearly establish whether Edition_No is globally unique, so assume it is a partial key under Book. Relationships: Writes (Author–Book; source says each book has an author, but it does not state whether a book may have multiple authors—assume M:N if co-authorship is supported); Published_By (Publisher–Book, one publisher per book under the text); Has (Book–Edition, 1:N with Edition weak under Book); Reviews between an Author and Book, with Rating and Feedback descriptive attributes. Example mapping: Book(ISBN PK,Title,Price,Year,PageCount,AuthorID FK if exactly one author,PubID FK); BookCategory(ISBN,Category); Author(A_ID PK,Name); AuthorPhone(A_ID,Phone); Publisher(PubID PK,Name,Email); PublisherPhone(PubID,PhoneNo); Edition(ISBN,Edition_No,Type) with PK (ISBN,Edition_No); Review(A_ID,ISBN,Rating,Feedback) with PK (A_ID,ISBN) unless multiple reviews are allowed.

ER-to-schema conversion examples shown on source PDF pages 82–90 and 91–94.

• Professor–Course 1:N example (p.82): Professor(Name, Phone, Address, Sex, CourseID); Course(CourseID, Name, Credits); Professor.CourseID FK references Course.CourseID. • Professor–Department 1:N example (p.83): Professor(Name, Phone, Address, Sex, Dep_Name); Department(Dep_Name, Location); Dep_Name FK references Department. • Student–Course M:N example (p.84): Student(RollNo PK, Name, Address); Course(CourseID PK, Name, Credits); Enrolls(RollNo FK, CourseID FK), composite PK (RollNo, CourseID). • Recursive Employee (p.85): Employee(Employee_ID PK, Name, Manager_ID FK → Employee.Employee_ID). • Professor–SponsoredProject M:N with attribute (p.86): Professor(EmpNo PK, Name, Address); SponsoredProject(Prj_ID PK, Name, Duration); WorksFor(EmpNo FK, Prj_ID FK, PercentTime), composite PK (EmpNo, Prj_ID) if one assignment per pair. • Employee–Dependents weak entity (p.87): Employee(EmpNo PK,EName); Medicare_Dependents(EmpNo FK, Depnd_Name, Relationship_with), PK (EmpNo, Depnd_Name). • Composite attribute example (p.88): flatten name/address components into columns. • Multivalued phone example (p.89): Inst_Phone(ID FK, Phone_Number), PK (ID, Phone_Number). • Banking schema example (p.91) and company-like schema (p.93) are addressed above and in the conversion section.

9. Quick Revision Sheet

Concept	Remember
Entity / entity set	A distinguishable real-world object / collection of same-type entities.
Attribute / domain	Descriptive property / permitted set of values.
Simple / composite	Atomic value / meaningful components (e.g., name → first, middle, last).
Single / multivalued	One value per entity / set of values (e.g., phone numbers).
Derived attribute	Computed from other attributes (e.g., age from date of birth); usually do not store redundantly.
Chen notation	Rectangle = entity; diamond = relationship; oval = attribute; underlined = key; double rectangle = weak entity; double diamond = identifying relationship; dashed underline = partial key; double line = total participation.
Relationship degree	Binary = 2 entity sets; ternary = 3; n-ary = n.
Cardinality	1:1, 1:N, N:1, M:N. Arrows/lines in the source notation indicate one/many constraints.
Minimum–maximum	l..h states minimum and maximum participation; 0..* = optional, any number; 1..1 = exactly one.
Total / partial	Total: every entity participates at least once. Partial: some may not participate. Not the same as weak/strong.
Keys	Super key uniquely identifies; candidate key is minimal super key; primary key is selected candidate key; FK references another relation's key.
Relationship key	Usually the union of participating entity keys; include descriptive attributes when needed to distinguish repeated relationship instances.
Strong / weak	Strong has own key; weak depends on owner and uses owner PK + partial key.
Strong entity mapping	Create a relation with its attributes; underline/mark PK.
1:N mapping	Place the one-side PK as FK on the many-side relation.
M:N mapping	Create a separate relation with both PKs as FKs plus relationship attributes.
1:1 mapping	Place FK on either side, guided by participation and avoiding unnecessary nulls.
Weak entity mapping	Include owner PK and discriminator in weak entity relation; this usually makes separate identifying-relationship relation redundant.
Composite / multivalued mapping	Flatten composite components into columns; make a separate relation for each multivalued attribute with owner PK + value.
Schema-to-ER inference	PKs identify entities; FKs indicate links; a separate relation containing two entity FKs often indicates M:N. Schema alone may not reveal minimum participation.

Final self-check before an exam: underline every PK; mark every FK; decide cardinality separately from participation; identify whether relationship attributes depend on the association; for weak entities, combine owner PK with the partial key; for M:N, create a bridge relation; never infer total participation solely from the presence of a foreign key.